Interacting Multipoles of Rank 1 and 3

$$-\frac{1}{15}\cdot T_{\alpha\beta\gamma\delta}\mu_{\alpha}^{B}\Omega_{\beta\gamma\delta}^{A}=R_z^{-9}\cdot-\frac{1}{15}\cdot\begin{bmatrix}+105\cdot R_\alpha\cdot R_\beta\cdot R_\gamma\cdot R_\delta\cdot\mu_\alpha^B\cdot\Omega_{\beta\gamma\delta}^A\\-15\cdot R_z^2\cdot R_\alpha\cdot R_\beta\cdot\delta\left(\gamma,\delta\right)\cdot\mu_\alpha^B\cdot\Omega_{\beta\gamma\delta}^A\\-15\cdot R_z^2\cdot R_\alpha\cdot R_\gamma\cdot\delta\left(\beta,\delta\right)\cdot\mu_\alpha^B\cdot\Omega_{\beta\gamma\delta}^A\\-15\cdot R_z^2\cdot R_\alpha\cdot R_\delta\cdot\delta\left(\beta,\gamma\right)\cdot\mu_\alpha^B\cdot\Omega_{\beta\gamma\delta}^A\\-15\cdot R_z^2\cdot R_\beta\cdot R_\gamma\cdot\delta\left(\alpha,\delta\right)\cdot\mu_\alpha^B\cdot\Omega_{\beta\gamma\delta}^A\\-15\cdot R_z^2\cdot R_\beta\cdot R_\delta\cdot\delta\left(\alpha,\gamma\right)\cdot\mu_\alpha^B\cdot\Omega_{\beta\gamma\delta}^A\\-15\cdot R_z^2\cdot R_\gamma\cdot R_\delta\cdot\delta\left(\alpha,\beta\right)\cdot\mu_\alpha^B\cdot\Omega_{\beta\gamma\delta}^A\\+3\cdot R_z^4\cdot\delta\left(\alpha,\beta\right)\cdot\delta\left(\gamma,\delta\right)\cdot\mu_\alpha^B\cdot\Omega_{\beta\gamma\delta}^A\\+3\cdot R_z^4\cdot\delta\left(\alpha,\gamma\right)\cdot\delta\left(\beta,\delta\right)\cdot\mu_\alpha^B\cdot\Omega_{\beta\gamma\delta}^A\\+3\cdot R_z^4\cdot\delta\left(\alpha,\delta\right)\cdot\delta\left(\beta,\gamma\right)\cdot\mu_\alpha^B\cdot\Omega_{\beta\gamma\delta}^A\end{bmatrix}\tag{1}$$

Simplify deltas:

$$-\frac{1}{15}\cdot T_{\alpha\beta\gamma\delta}\mu_{\alpha}^{B}\Omega_{\beta\gamma\delta}^{A}=R_z^{-9}\cdot-\frac{1}{15}\cdot\begin{bmatrix}+105\cdot R_\alpha\cdot R_\beta\cdot R_\gamma\cdot R_\delta\cdot\mu_\alpha^B\cdot\Omega_{\beta\gamma\delta}^A\\-15\cdot R_z^2\cdot R_\alpha\cdot R_\beta\cdot\delta\left(\gamma,\gamma\right)\cdot\mu_\alpha^B\cdot\Omega_{\beta\gamma\gamma}^A\\-15\cdot R_z^2\cdot R_\alpha\cdot R_\gamma\cdot\delta\left(\beta,\beta\right)\cdot\mu_\alpha^B\cdot\Omega_{\beta\beta\gamma}^A\\-15\cdot R_z^2\cdot R_\alpha\cdot R_\delta\cdot\delta\left(\beta,\beta\right)\cdot\mu_\alpha^B\cdot\Omega_{\beta\beta\delta}^A\\-15\cdot R_z^2\cdot R_\beta\cdot R_\gamma\cdot\delta\left(\alpha,\alpha\right)\cdot\mu_\alpha^B\cdot\Omega_{\alpha\beta\gamma}^A\\-15\cdot R_z^2\cdot R_\beta\cdot R_\delta\cdot\delta\left(\alpha,\alpha\right)\cdot\mu_\alpha^B\cdot\Omega_{\alpha\beta\delta}^A\\-15\cdot R_z^2\cdot R_\gamma\cdot R_\delta\cdot\delta\left(\alpha,\alpha\right)\cdot\mu_\alpha^B\cdot\Omega_{\alpha\gamma\delta}^A\\+3\cdot R_z^4\cdot\delta\left(\alpha,\alpha\right)\cdot\delta\left(\beta,\beta\right)\cdot\mu_\alpha^B\cdot\Omega_{\alpha\beta\beta}^A\\+3\cdot R_z^4\cdot\delta\left(\alpha,\alpha\right)\cdot\delta\left(\beta,\beta\right)\cdot\mu_\alpha^B\cdot\Omega_{\alpha\beta\beta}^A\\+3\cdot R_z^4\cdot\delta\left(\alpha,\alpha\right)\cdot\delta\left(\beta,\beta\right)\cdot\mu_\alpha^B\cdot\Omega_{\alpha\beta\beta}^A\end{bmatrix}\tag{2a}$$

$$\xRightarrow{\text{cleaned up}}R_z^{-9}\cdot-\frac{1}{15}\cdot\begin{bmatrix}+105\cdot R_\alpha\cdot R_\beta\cdot R_\gamma\cdot R_\delta\cdot\mu_\alpha^B\cdot\Omega_{\beta\gamma\delta}^A\\-15\cdot R_z^2\cdot R_\alpha\cdot R_\beta\cdot\mu_\alpha^B\cdot\Omega_{\beta\gamma\gamma}^A\\-15\cdot R_z^2\cdot R_\alpha\cdot R_\gamma\cdot\mu_\alpha^B\cdot\Omega_{\beta\beta\gamma}^A\\-15\cdot R_z^2\cdot R_\alpha\cdot R_\delta\cdot\mu_\alpha^B\cdot\Omega_{\beta\beta\delta}^A\\-15\cdot R_z^2\cdot R_\beta\cdot R_\gamma\cdot\mu_\alpha^B\cdot\Omega_{\alpha\beta\gamma}^A\\-15\cdot R_z^2\cdot R_\beta\cdot R_\delta\cdot\mu_\alpha^B\cdot\Omega_{\alpha\beta\delta}^A\\-15\cdot R_z^2\cdot R_\gamma\cdot R_\delta\cdot\mu_\alpha^B\cdot\Omega_{\alpha\gamma\delta}^A\\+3\cdot R_z^4\cdot\mu_\alpha^B\cdot\Omega_{\alpha\beta\beta}^A\\+3\cdot R_z^4\cdot\mu_\alpha^B\cdot\Omega_{\alpha\beta\beta}^A\\+3\cdot R_z^4\cdot\mu_\alpha^B\cdot\Omega_{\alpha\beta\beta}^A\end{bmatrix}\tag{2b}$$

$$\xRightarrow{\text{reduced indices}}R_z^{-9}\cdot-\frac{1}{15}\cdot\begin{bmatrix}+105\cdot R_\alpha\cdot R_\beta\cdot R_\gamma\cdot R_\delta\cdot\mu_\alpha^B\cdot\Omega_{\beta\gamma\delta}^A\\-15\cdot R_z^2\cdot R_\alpha\cdot R_\beta\cdot\mu_\alpha^B\cdot\Omega_{\beta\gamma\gamma}^A\\-15\cdot R_z^2\cdot R_\alpha\cdot R_\gamma\cdot\mu_\alpha^B\cdot\Omega_{\beta\beta\gamma}^A\\-15\cdot R_z^2\cdot R_\alpha\cdot R_\gamma\cdot\mu_\alpha^B\cdot\Omega_{\beta\beta\gamma}^A\\-15\cdot R_z^2\cdot R_\beta\cdot R_\gamma\cdot\mu_\alpha^B\cdot\Omega_{\alpha\beta\gamma}^A\\-15\cdot R_z^2\cdot R_\beta\cdot R_\gamma\cdot\mu_\alpha^B\cdot\Omega_{\alpha\beta\gamma}^A\\-15\cdot R_z^2\cdot R_\gamma\cdot R_\beta\cdot\mu_\alpha^B\cdot\Omega_{\alpha\beta\gamma}^A\\+3\cdot R_z^4\cdot\mu_\alpha^B\cdot\Omega_{\alpha\beta\beta}^A\\+3\cdot R_z^4\cdot\mu_\alpha^B\cdot\Omega_{\alpha\beta\beta}^A\\+3\cdot R_z^4\cdot\mu_\alpha^B\cdot\Omega_{\alpha\beta\beta}^A\end{bmatrix}\tag{2c}$$

$$\xRightarrow{\text{added up}}R_z^{-9}\cdot-\frac{1}{15}\cdot\begin{bmatrix}+105\cdot R_\alpha\cdot R_\beta\cdot R_\gamma\cdot R_\delta\cdot\mu_\alpha^B\cdot\Omega_{\beta\gamma\delta}^A\\-15\cdot R_z^2\cdot R_\alpha\cdot R_\beta\cdot\mu_\alpha^B\cdot\Omega_{\beta\gamma\gamma}^A\\-30\cdot R_z^2\cdot R_\alpha\cdot R_\gamma\cdot\mu_\alpha^B\cdot\Omega_{\beta\beta\gamma}^A\\-45\cdot R_z^2\cdot R_\beta\cdot R_\gamma\cdot\mu_\alpha^B\cdot\Omega_{\alpha\beta\gamma}^A\\+9\cdot R_z^4\cdot\mu_\alpha^B\cdot\Omega_{\alpha\beta\beta}^A\end{bmatrix}\tag{2d}$$

Simplify R:

$$-\frac{1}{15} \cdot T_{\alpha\beta\gamma\delta} \mu_\alpha^B \Omega_{\beta\gamma\delta}^A = R_z^{-9} \cdot -\frac{1}{15} \cdot \begin{bmatrix} +105 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \mu_z^B \cdot \Omega_{zzz}^A \\ -15 \cdot R_z^2 \cdot R_z \cdot R_z \cdot \mu_z^B \cdot \Omega_{z\gamma\gamma}^A \\ -30 \cdot R_z^2 \cdot R_z \cdot R_z \cdot \mu_z^B \cdot \Omega_{z\beta\beta}^A \\ -45 \cdot R_z^2 \cdot R_z \cdot R_z \cdot \mu_\alpha^B \cdot \Omega_{zz\alpha}^A \\ +9 \cdot R_z^4 \cdot \mu_\alpha^B \cdot \Omega_{\alpha\beta\beta}^A \end{bmatrix} \tag{3a}$$

$$\xRightarrow{\text{cleaned up}} R_z^{-9} \cdot -\frac{1}{15} \cdot \begin{bmatrix} -45 \cdot R_z^4 \cdot \mu_\alpha^B \cdot \Omega_{zz\alpha}^A \\ +9 \cdot R_z^4 \cdot \mu_\alpha^B \cdot \Omega_{\alpha\beta\beta}^A \end{bmatrix} \tag{3b}$$

Simplify Dipoles:

$$-\frac{1}{15} \cdot T_{\alpha\beta\gamma\delta} \mu_\alpha^B \Omega_{\beta\gamma\delta}^A = R_z^{-9} \cdot -\frac{1}{15} \cdot \begin{bmatrix} -45 \cdot R_z^4 \cdot \mu_y^B \cdot \Omega_{yzz}^A \\ +9 \cdot R_z^4 \cdot \mu_y^B \cdot \Omega_{y\beta\beta}^A \end{bmatrix} \tag{4a}$$

$$\xRightarrow{\text{reduced indices}} R_z^{-9} \cdot -\frac{1}{15} \cdot \begin{bmatrix} -45 \cdot R_z^4 \cdot \mu_y^B \cdot \Omega_{yzz}^A \\ +9 \cdot R_z^4 \cdot \mu_y^B \cdot \Omega_{y\alpha\alpha}^A \end{bmatrix} \tag{4b}$$

Exploit Traceless Conditions:

$$-\frac{1}{15} \cdot T_{\alpha\beta\gamma\delta} \mu_\alpha^B \Omega_{\beta\gamma\delta}^A = R_z^{-9} \cdot -\frac{1}{15} \cdot \begin{bmatrix} -45 \cdot R_z^4 \cdot \mu_y^B \cdot \Omega_{yzz}^A \end{bmatrix} \tag{5}$$

Combining everything:

$$-\frac{1}{15} \cdot T_{\alpha\beta\gamma\delta} \mu_\alpha^B \Omega_{\beta\gamma\delta}^A = \begin{bmatrix} +3 \cdot R_z^{-5} \cdot \mu_y^B \cdot \Omega_{yzz}^A \end{bmatrix} \tag{6}$$